so on through a great succession of towers and wires until the request reaches one of the true masters of time, a Network Time Server run by or referenced against the atomic clocks kept at places like the National Institute of Standards and Technology in the United States, the Federal Institute of Meteorology and Climatology in Switzerland, and the National Institute of Information and Communications Technology in Japan. That long invisible journey, accomplished in a fraction of a second, is why you don't see a blinking 12:00 on your phone's screen every time you power it up again after its battery runs out.

I was born in 1983, at the end of the world in which people set the time for themselves. That was the year that the US Department of Defense split its internal system of interconnected computers in half, creating one network for the use of the defense establishment, called MILNET, and another network for the public, called the Internet. Before the year was out, new rules defined the boundaries of this virtual space, giving rise to the Domain Name System that we still use today—the.govs, .mils,.edus, and, of course,.coms—and the country codes assigned to the rest of the world:.uk, .de, .fr, .cn, .ru, and so on. Already, my country (and so I) had an advantage, an edge. And yet it would be another six years before the World Wide Web was invented, and about nine years before my family got a computer with a modem that could connect to it.

Of course, the Internet is not a single entity, although we tend to refer to it as if it were. The technical reality is that there are new networks born every day on the global cluster of interconnected communications networks that you—and about three billion other people, or roughly 42 percent of the world's population—use regularly. Still, I'm going to use the term in its broadest sense, to mean the universal network of networks connecting the majority of the world's computers to one another via a set of shared protocols.

Some of you may worry that you don't know a protocol from a hole in the wall, but all of us have made use of many. Think of protocols as languages for machines, the common rules they follow to be understood by one another. If you're around my age, you might remember having to type the "http" at the beginning of a website's address into the address bar of your Web browser. This refers to the Hypertext Transfer Protocol, the language you use to access the World Wide Web, that massive collection of mostly text-based but also audio- and video-capable sites like Google and YouTube and Facebook. Every time you check your email, you use a language like IMAP (Internet Message Access Protocol), SMTP (Simple Mail Transfer Protocol), or POP3 (Post Office Protocol). File transfers pass through the Internet using FTP (File Transfer